PROJECT REPORT

Molecular Docking and ADMET Prediction of Berberine Against Type 2 Diabetes Mellitus Target

Submitted by

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Abstract

Type 2 Diabetes Mellitus (T2DM) is a chronic metabolic disorder aligned by insulin resistance and impaired glucose homeostasis, which is a major global health challenge. Natural products have gained increasing importance as potential therapeutic agents due to their diverse biological activities and favourable safety profiles. Berberine, a bioactive isoquinoline alkaloid isolated from several medicinal plants, has showed promising antidiabetic properties. The present study taken an integrated computational approach to find the therapeutic potential of berberine against T2DM-associated molecular targets.

Potential targets of berberine and T2DM-related genes were collected from publicly available databases, and common targets were identified for subsequent network pharmacology analysis. Protein-protein interaction (PPI) network analysis was performed using STRING, followed by KEGG pathway enrichment to identify key biological pathways involved in diabetic progression. Peroxisome Proliferator-Activated Receptor Gamma (PPARG) was selected as the principal therapeutic target due to its central role in insulin sensitivity and glucose metabolism.

The binding interaction between berberine and PPARG was evaluated through molecular docking using AutoDock Vina, followed by validation using the CB-Dock2 web server, which employs the AutoDock Vina docking engine. The docking results indicated a favourable binding affinity and stable protein–ligand interactions. Drug-likeness and pharmacokinetic properties were assessed using SwissADME, revealing high gastrointestinal absorption, acceptable lipophilicity, and compliance with Lipinski's Rule of Five. Toxicity assessment using ProTox-3.0 suggested an acceptable safety profile with no predicted hepatotoxicity, nephrotoxicity, or cardiotoxicity.

Overall, the integrated network pharmacology, molecular docking, ADMET, and toxicity analyses suggest that berberine is a promising natural

therapeutic agent for managing Type 2 Diabetes Mellitus through modulation of PPARG and associated insulin signalling pathways.

1. Introduction

1.1 Type 2 Diabetes Mellitus

Type 2 Diabetes Mellitus (T2DM) is a chronic metabolic disorder characterized by persistent hyperglycemia resulting from insulin resistance and impaired insulin secretion. It is one of the most common non-communicable diseases worldwide and represents a major public health challenge due to its increasing incidence and associated complications. According to the International Diabetes Federation (IDF), hundreds of millions of people are currently living with diabetes, with Type 2 Diabetes accounting for approximately 90-95% of all diagnosed cases. The disease is closely associated with obesity, sedentary lifestyle, unhealthy dietary habits, genetic predisposition, and aging.

The long-term progression of T2DM contributes to both microvascular and macrovascular complications, including diabetic nephropathy, retinopathy, neuropathy, cardiovascular disease, and stroke. Although there are several pharmacological agents for the treatment of T2DM, many are associated with adverse effects, reduced efficacy over time, and high treatment costs. Therefore, the identification of safer and more effective therapeutic agents remains an important objective in diabetic research.

1.2 Berberine as a Potential Antidiabetic Compound

Natural products have been important sources for drug discovery due to their structural diversity and wide range of biological activities. Berberine is an isoquinoline alkaloid that is found naturally in medicinal plants like Berberis vulgaris, Coptis chinensis, and Hydrastis canadensis. Traditionally, berberine has been used in herbal medicine for treating gastrointestinal infections and inflammatory disorders. More recently, several experimental and clinical studies have demonstrated its beneficial effects on metabolic disorders, mainly Type 2 Diabetes Mellitus.

Berberine has been reported to improve insulin sensitivity, regulate glucose metabolism, reduce hepatic glucose production, enhance glucose uptake, and modulate lipid metabolism.

1.3 Role of PPARG in Type 2 Diabetes Mellitus

Peroxisome Proliferator-Activated Receptor Gamma (PPARG) is a ligand-activated nuclear receptor that functions as a transcription factor regulating genes involved in glucose homeostasis, lipid metabolism, adipocyte differentiation, and insulin sensitivity. PPARG plays a main role in maintaining metabolic balance by enhancing insulin responsiveness and promoting glucose utilization in peripheral tissues.

Several clinically approved antidiabetic drugs, including thiazolidinediones, exert their therapeutic effects through activation of PPARG. PPARG has emerged as one of the most extensively studied molecular targets for the treatment of T2DM. Examining the interaction between berberine and PPARG may therefore provide valuable insights into the molecular mechanisms underlying the antidiabetic activity of this natural compound.

1.4 Computational Drug Discovery

Computational drug discovery has emerged as an indispensable component of modern medicinal chemistry by enabling the rapid identification, optimization, and evaluation of potential drug candidates through computer-aided approaches. Unlike conventional experimental screening, computational techniques significantly reduce the time, cost, and resources required during the early stages of drug development while providing molecular-level insights into drug–target interactions.

Computer-Aided Drug Design (CADD) integrates principles of computational chemistry, structural biology, molecular modelling, and cheminformatics to investigate the interaction between small-molecule ligands and biological macromolecules. Among these techniques, molecular docking is one of the most widely employed structure-based drug design approaches for predicting the preferred binding orientation, binding affinity, and intermolecular interactions between a ligand and its target protein. These predictions provide valuable information regarding the stability of protein–ligand complexes and assist in identifying promising therapeutic candidates prior to experimental validation.

In addition to molecular docking, drug-likeness, pharmacokinetic, and toxicity evaluations have become integral components of computational drug discovery. In silico ADMET (Absorption, Distribution, Metabolism, Excretion, and Toxicity) prediction enables early assessment of pharmacological properties, thereby reducing the likelihood of late-stage drug failure. Molecular visualization tools further facilitate the detailed analysis of hydrogen bonding, hydrophobic interactions, and other non-covalent interactions responsible for ligand binding.

In the present study, berberine was investigated using an integrated computational workflow involving target identification, molecular docking with AutoDock Vina and CB-Dock2, protein–ligand interaction analysis, and in silico ADMET and toxicity prediction. This comprehensive computational strategy provides a systematic evaluation of the therapeutic potential of berberine against Peroxisome Proliferator-Activated Receptor Gamma (PPARG) as a potential target for the management of Type 2 Diabetes Mellitus.

2. Objectives

The primary objective of this study was to investigate the therapeutic potential of berberine against Type 2 Diabetes Mellitus (T2DM) using an integrated computational drug discovery approach.

The specific objectives of the study were:

- To identify the common therapeutic targets between berberine and Type 2 Diabetes Mellitus using network pharmacology.
- To construct and analyse the protein–protein interaction (PPI) network and identify significant biological pathways associated with T2DM.
- To identify Peroxisome Proliferator-Activated Receptor Gamma (PPARG) as a potential therapeutic target through KEGG pathway enrichment analysis.

- To evaluate the binding affinity and molecular interactions between berberine and PPARG using molecular docking.
- To assess the drug-likeness, pharmacokinetic (ADMET), and toxicity profiles of berberine using in silico prediction tools.

3. Materials and Methods

3.1 Overall Workflow

The present study employed an integrated computational drug discovery workflow comprising literature survey, ligand preparation, target identification, network pharmacology, protein preparation, molecular docking, protein–ligand interaction analysis, and *in silico* ADMET and toxicity prediction. The workflow was designed to systematically evaluate the therapeutic potential of berberine against Type 2 Diabetes Mellitus (T2DM). A schematic representation of the complete methodology adopted in this study is presented in Figure 1.

Figure 1. Overall computational workflow employed for molecular docking and ADMET prediction of berberine against Type 2 Diabetes Mellitus using an integrated *in silico* approach

3.2 Literature Survey

A comprehensive literature survey was conducted to establish the scientific background of the present study. Relevant peer-reviewed research articles, review papers, and publicly available scientific resources related to Type 2 Diabetes Mellitus (T2DM), berberine, Peroxisome Proliferator-Activated Receptor Gamma (PPARG), network pharmacology, molecular docking, and computational drug discovery were systematically reviewed. The literature survey facilitated the understanding of disease pathology, molecular mechanisms, therapeutic targets, and the pharmacological properties of berberine.

Scientific databases including PubMed, Google Scholar, ScienceDirect, SpringerLink, and Web of Science were consulted to obtain relevant and up-to-date literature. The information gathered from these sources was utilized to formulate the study design, select appropriate computational tools, and interpret the results obtained during the investigation.

3.3 Ligand Retrieval and Preparation

The chemical structure of berberine was retrieved from the PubChem database (https://pubchem.ncbi.nlm.nih.gov/) in SMILES (Simplified Molecular Input Line Entry System) format. The retrieved SMILES notation was converted into a three-dimensional (3D) molecular structure using Open Babel, which facilitated subsequent computational analysis.

The generated 3D ligand structure was prepared for molecular docking using AutoDockTools (MGLTools). During ligand preparation, polar hydrogen atoms were added, Gasteiger partial charges were assigned, and appropriate rotatable bonds were defined. The prepared ligand was subsequently saved in PDBQT format, which is required for molecular docking using the AutoDock Vina docking engine.

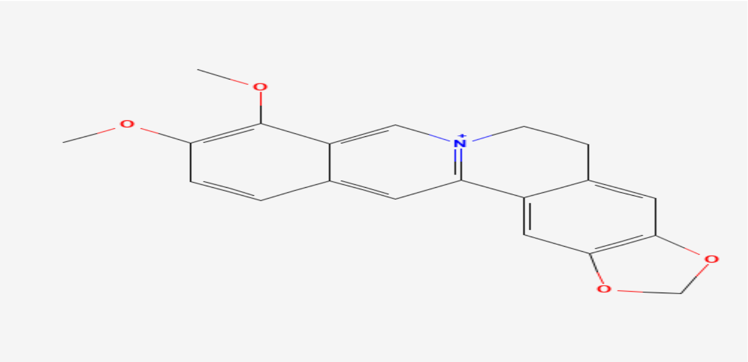

Figure 2. Two-dimensional chemical structure of berberine retrieved from PubChem database

3.4 Target Prediction

Potential molecular targets of berberine were predicted using the SwissTargetPrediction web server (http://www.swisstargetprediction.ch/). The canonical SMILES representation of berberine, obtained from the PubChem database, was used as the input. The target organism was specified as Homo sapiens, and the predicted protein targets were retrieved based on the chemical similarity of berberine to compounds with experimentally validated biological targets. The predicted targets were collected and used for subsequent network pharmacology analysis.

3.5 Disease-Associated Gene Collection

Genes associated with Type 2 Diabetes Mellitus (T2DM) were retrieved from the GeneCards database (https://www.genecards.org/), an integrated human gene database that provides comprehensive information on gene functions and disease associations. The keyword "Type 2 Diabetes Mellitus" was used to search for disease-related genes. Genes with high relevance scores and strong associations with T2DM were collected and used for subsequent network pharmacology analysis. The retrieved disease-associated genes were later compared with the predicted targets of berberine to identify common therapeutic targets.

3.6 Identification of Common Therapeutic Targets

The predicted molecular targets of berberine obtained from SwissTargetPrediction were compared with the Type 2 Diabetes Mellitus (T2DM)-associated genes retrieved from the GeneCards database to identify common therapeutic targets. The comparison was performed using Venny 2.1 (or specify the exact Venn analysis tool you used, e.g., InteractiVenn), which enables visualization of overlapping gene sets through a Venn diagram.

The common targets identified from the overlap analysis were considered potential therapeutic targets of berberine against T2DM and were selected for subsequent protein–protein interaction (PPI) network construction, KEGG pathway enrichment analysis, and molecular docking studies.

3.7 Protein–Protein Interaction (PPI) Network Construction

The common therapeutic targets identified from the overlap analysis were subjected to Protein–Protein Interaction (PPI) network analysis using the STRING database (Search Tool for the Retrieval of Interacting Genes/Proteins; https://string-db.org/). The target organism was specified as Homo sapiens, and interactions supported by experimental evidence, curated databases, co-expression, and text mining were considered for network construction.

The generated PPI network was exported for further topological analysis and visualization using Cytoscape. The PPI network facilitated the identification of highly interconnected proteins that may play significant roles in the molecular mechanism of berberine against Type 2 Diabetes Mellitus.

3.8 Network Visualization and Topological Analysis

The protein–protein interaction (PPI) network generated using the STRING database was imported into Cytoscape (Version 3.x) for network visualization and topological analysis. Cytoscape was employed to visualize the interaction network and facilitate the identification of highly connected proteins within the network.

Topological parameters, including degree centrality, were analyzed to identify hub proteins that may play significant roles in the molecular mechanism of berberine against Type 2 Diabetes Mellitus. The identified hub proteins were subsequently considered for pathway enrichment analysis and molecular docking studies.

3.9 KEGG Pathway Enrichment Analysis

The common therapeutic targets identified through overlap analysis were subjected to Kyoto Encyclopaedia of Genes and Genomes (KEGG) pathway enrichment analysis using the STRING database (Version 12.0) (https://string-db.org/). The analysis was performed with Homo sapiens selected as the target organism. KEGG pathway enrichment analysis was carried out to identify significantly enriched biological pathways associated with the common therapeutic targets and to investigate the molecular mechanisms through which berberine may exert its therapeutic effects against Type 2 Diabetes Mellitus (T2DM).

The enriched KEGG pathways were evaluated based on their statistical significance and biological relevance. The pathway enrichment results were subsequently used to support the selection of the target protein for molecular docking studies.

3.10 Protein Retrieval and Preparation

The three-dimensional crystal structure of Peroxisome Proliferator-Activated Receptor Gamma (PPARG) was retrieved from the RCSB Protein Data Bank (RCSB PDB) (https://www.rcsb.org/) in PDB format. PPARG was selected as the target protein based on the results of the KEGG pathway enrichment analysis and its established role in regulating glucose homeostasis, lipid metabolism, and insulin sensitivity in Type 2 Diabetes Mellitus (T2DM).

The retrieved protein structure was prepared for molecular docking using PyMOL and AutoDockTools (MGLTools). During protein preparation, water molecules, co-crystallized ligands, and other non-essential heteroatoms were removed, while polar hydrogen atoms and Kollman united atom charges were added. The prepared protein structure was subsequently saved in PDBQT format for molecular docking using the AutoDock Vina docking engine.

3.11 Molecular Docking

Molecular docking was performed to investigate the binding interaction between berberine and Peroxisome Proliferator-Activated Receptor Gamma (PPARG) using AutoDock Vina (Version 1.2.x). The prepared protein and ligand structures in PDBQT format were used as input for the docking simulations. A docking grid box was defined to encompass the active binding site of PPARG, allowing the ligand to explore potential binding conformations within the receptor.

AutoDock Vina employs a gradient-based conformational search algorithm coupled with an empirical scoring function to predict ligand binding poses and estimate binding affinities. Multiple docking poses were generated, and the binding conformations were ranked according to their predicted binding energies (kcal/mol). The docking pose exhibiting the lowest binding energy was selected for subsequent protein–ligand interaction analysis.

3.12 Protein–Ligand Interaction Analysis

The molecular docking results were analyzed to examine the interactions between berberine and the active site residues of Peroxisome Proliferator-Activated Receptor Gamma (PPARG). The selected docking complex exhibiting the most favorable binding energy was visualized using PyMOL (Version 3.x) to generate the three-dimensional (3D) representation of the protein–ligand complex.

Detailed two-dimensional (2D) interaction analysis was performed using BIOVIA Discovery Studio Visualizer (Version 2024). The analysis was conducted to identify the amino acid residues involved in ligand binding and characterize the intermolecular interactions, including hydrogen bonds, hydrophobic interactions, π–π interactions, π–alkyl interactions, van der Waals interactions, and other non-covalent interactions contributing to the stability of the protein–ligand complex.

3.13 ADMET Prediction

The pharmacokinetic and drug-likeness properties of berberine were evaluated using the SwissADME web server (https://www.swissadme.ch/). The SMILES representation of berberine, obtained from the PubChem database, was used as the input for the analysis. SwissADME was employed to predict the physicochemical properties, lipophilicity, water solubility, pharmacokinetic parameters, drug-likeness, and medicinal chemistry characteristics of berberine.

The analysis included the evaluation of important parameters such as molecular weight, topological polar surface area (TPSA), hydrogen bond donors and acceptors, lipophilicity (LogP), gastrointestinal (GI) absorption, blood–brain barrier (BBB) permeability, P-glycoprotein (P-gp) substrate prediction, cytochrome P450 (CYP450) enzyme inhibition, and compliance with Lipinski's Rule of Five. These predictions were used to assess the suitability of berberine as a potential orally active therapeutic candidate.

3.14 Toxicity Prediction

The toxicity profile of berberine was predicted using the ProTox-3.0 web server (https://tox.charite.de/protox3/). The SMILES representation of berberine, obtained from the PubChem database, was submitted to the server for in silico toxicity assessment. ProTox-3.0 integrates machine learning-based prediction models with molecular similarity approaches to estimate the toxicological properties of chemical compounds.

The toxicity assessment included the prediction of oral toxicity (LD_{50}), toxicity class, hepatotoxicity, carcinogenicity, mutagenicity, cytotoxicity, immunotoxicity, and other relevant toxicological endpoints. The predicted toxicity profile was used to evaluate the safety of berberine as a potential therapeutic candidate for the treatment of Type 2 Diabetes Mellitus (T2DM).

4. Results and Discussion

4.1 Target Prediction of Berberine

SwissTargetPrediction identified 22 potential molecular targets for **berberine** based on its chemical structure and similarity to compounds with experimentally validated biological activities. These predicted targets belong to diverse protein families, including enzymes, kinases, nuclear receptors, membrane receptors, and transport proteins, indicating that berberine may exert its pharmacological effects through a multi-target mechanism.

The predicted targets are involved in several biological processes associated with glucose metabolism, insulin signalling, lipid metabolism, inflammation, and energy homeostasis, which are closely related to the pathophysiology of Type 2 Diabetes Mellitus (T2DM). The complete list of predicted molecular targets obtained from SwissTargetPrediction is presented in Table 1.

Table 1. Predicted molecular targets of berberine obtained using SwissTargetPrediction.

S.No.	Gene Symbol	Protein Name	Target Class
1	PPARG	Peroxisome proliferator-activated receptor gamma	Nuclear receptor
2	PPARA	Peroxisome proliferator-activated receptor alpha	Nuclear receptor
3	AKT1	RAC-alpha serine/threonine-protein kinase	Kinase
4	AKT2	RAC-beta serine/threonine-protein kinase	Kinase
5	INSR	Insulin receptor	Receptor tyrosine kinase
6	IRS1	Insulin receptor substrate 1	Signalling adaptor protein
7	IRS2	Insulin receptor substrate 2	Signalling adaptor protein
8	PIK3CA	Phosphatidylinositol 3-kinase catalytic subunit alpha	Lipid kinase
9	PIK3R1	Phosphatidylinositol 3-kinase regulatory subunit alpha	Regulatory protein
10	MTOR	Mechanistic target of rapamycin kinase	Serine/threonine kinase
11	FOXO1	Forkhead box protein O1	Transcription factor
12	SIRT1	NAD-dependent protein deacetylase SIRT1	Enzyme (Deacetylase)
13	MAPK1	Mitogen-activated protein kinase 1	Kinase
14	MAPK3	Mitogen-activated protein kinase 3	Kinase
15	STAT3	Signal transducer and activator of transcription 3	Transcription factor
16	TNF	Tumour necrosis factor	Cytokine
17	CXCL8	C-X-C motif chemokine ligand 8	Chemokine
18	GSK3B	Glycogen synthase kinase-3 beta	Kinase
19	ADIPOQ	Adiponectin	Hormone (Adipokine)
20	PRKAA1	AMP-activated protein kinase catalytic subunit alpha-1	Kinase
21	PRKAA2	AMP-activated protein kinase catalytic subunit alpha-2	Kinase
22	JAK2	Janus kinase 2	Tyrosine kinase

4.2 Identification of Common Therapeutic Targets

To identify the potential therapeutic targets of berberine against Type 2 Diabetes Mellitus (T2DM), the 22 predicted molecular targets obtained from SwissTargetPrediction were compared with 6,452 T2DM-associated genes retrieved from the GeneCards database. The overlap analysis identified 19 common therapeutic targets, while 3 predicted targets were unique to berberine and 6,433 genes were unique to T2DM. The identified overlapping genes were considered potential therapeutic targets and selected for subsequent protein–protein interaction (PPI) network construction, KEGG pathway enrichment analysis, and molecular docking studies.

The identified overlapping genes participate in several biological processes associated with insulin signalling, glucose homeostasis, lipid metabolism, inflammatory responses, and cellular energy regulation. These common targets were subsequently used for protein–protein interaction (PPI) network construction, KEGG pathway enrichment analysis, and molecular docking studies.

The overlap between the predicted drug targets and disease-associated genes is illustrated in Figure 1, while the list of common therapeutic targets is presented in Table 2.

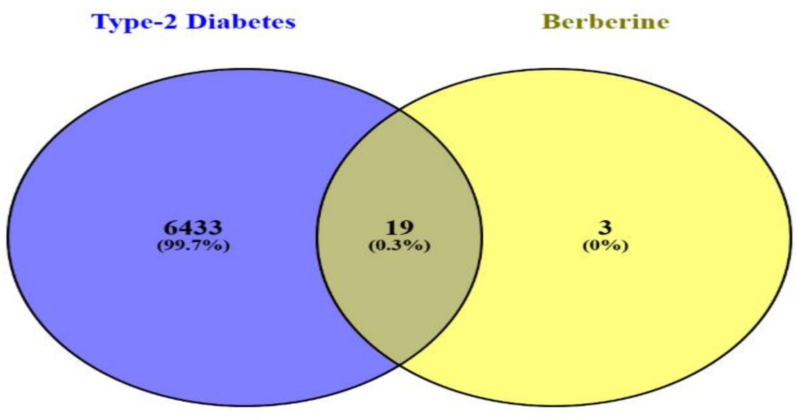

Figure 1. Venn diagram illustrating the overlap between predicted molecular targets of berberine obtained from SwissTargetPrediction and Type 2 Diabetes Mellitus-associated genes retrieved from the GeneCards database

S.No.	Gene Symbol	Protein Name
1	PPARG	Peroxisome proliferator-activated receptor gamma
2	PPARA	Peroxisome proliferator-activated receptor alpha
3	INSR	Insulin receptor
4	IRS1	Insulin receptor substrate 1
5	IRS2	Insulin receptor substrate 2
6	PIK3CA	PI3K catalytic subunit alpha
7	PIK3R1	PI3K regulatory subunit alpha
8	AKT1	RAC-alpha serine/threonine-protein kinase
9	AKT2	RAC-beta serine/threonine-protein kinase
10	MTOR	Mechanistic target of rapamycin
11	FOXO1	Forkhead box protein O1
12	SIRT1	Sirtuin 1
13	GSK3B	Glycogen synthase kinase-3 beta
14	MAPK1	Mitogen-activated protein kinase 1
15	MAPK3	Mitogen-activated protein kinase 3
16	STAT3	Signal transducer and activator of transcription 3
17	TNF	Tumour necrosis factor
18	CXCL8	C-X-C motif chemokine ligand 8
19	ADIPOQ	Adiponectin

Table 2. Common therapeutic targets of berberine against Type 2 Diabetes Mellitus identified through overlap analysis

4.3 Protein–Protein Interaction (PPI) Network Analysis

To investigate the functional relationships among the identified common therapeutic targets, a protein–protein interaction (PPI) network was constructed using the STRING database (Version 12.0). The 19 overlapping genes obtained from the Venn analysis were submitted to STRING using Homo sapiens as the reference organism. The resulting network consisted of 19 nodes and 152 interaction edges, with an average node degree of 16 and an average local clustering coefficient of 0.902. The expected number of edges for a random network of similar size was 49, whereas the observed network contained 152 edges, indicating a significantly higher degree of functional connectivity among the proteins. The PPI enrichment p-value ($< 1.0 \times 10^{-16}$) demonstrated that the observed interactions were highly significant and unlikely to have occurred by chance.

The interaction network comprised proteins involved in several biological processes associated with Type 2 Diabetes Mellitus (T2DM), including insulin signalling, glucose metabolism, lipid metabolism, inflammatory responses, and cellular energy homeostasis. The extensive connectivity among these proteins suggests that berberine may exert its therapeutic effects through the simultaneous modulation of multiple interconnected molecular targets.

The constructed PPI network is presented in Figure 2.

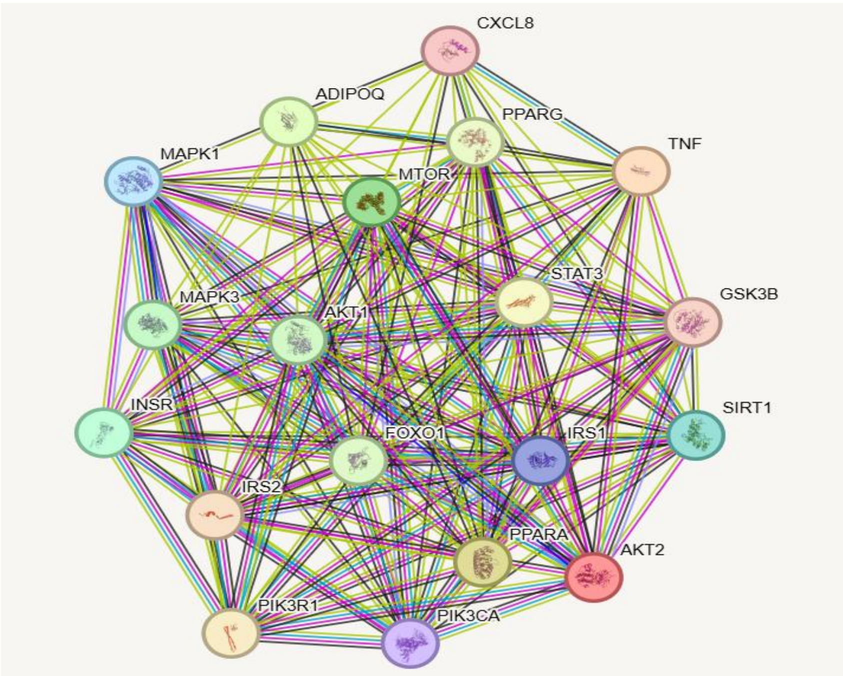

Figure 2. Protein–protein interaction (PPI) network of the 19 common therapeutic targets generated using the STRING database.

Discussion

The PPI network revealed a highly interconnected interaction pattern among the common therapeutic targets. Key metabolic regulators, including PPARG, AKT1, AKT2, INSR, IRS1, IRS2, PIK3CA, PIK3R1, MTOR, FOXO1, and SIRT1, exhibited extensive interactions with other network components, highlighting their important roles in insulin signalling, glucose homeostasis, and energy metabolism. Furthermore, inflammatory mediators such as TNF, CXCL8, and STAT3 were closely associated with metabolic signalling proteins, indicating a strong relationship between metabolic regulation and inflammatory processes in T2DM.

The significantly higher number of observed interactions compared to the expected interactions, together with the highly significant PPI enrichment p-value ($< 1.0 \times 10^{-16}$), confirms that these proteins are biologically connected rather than randomly associated. These findings support the hypothesis that berberine exerts its antidiabetic activity through a multi-target mechanism, influencing several interconnected signalling pathways involved in the pathogenesis of Type 2 Diabetes Mellitus.

4.4 KEGG Pathway Enrichment Analysis

To investigate the biological pathways associated with the identified common therapeutic targets, Kyoto Encyclopaedia of Genes and Genomes (KEGG) pathway enrichment analysis was performed using the STRING database. The enrichment analysis revealed that the overlapping targets were significantly involved in several metabolic and signalling pathways associated with the pathogenesis of Type 2 Diabetes Mellitus (T2DM). The enriched pathways primarily regulate insulin sensitivity, glucose metabolism, lipid metabolism, inflammatory responses, and cellular energy homeostasis. The top diabetes-related enriched pathways identified in this study are presented in Table 3.

KEGG ID	KEGG Pathway	Gene Count	FDR
hsa04930	Type II diabetes mellitus	10	6.08×10^{-20}
hsa04931	Insulin resistance	13	6.34×10^{-23}
hsa04910	Insulin signalling pathway	12	4.67×10^{-20}
hsa04152	AMPK signalling pathway	12	2.63×10^{-20}
hsa04068	FoxO signalling pathway	12	3.44×10^{-20}
hsa04150	mTOR signalling pathway	11	1.83×10^{-17}
hsa04920	Adipocytokine signalling pathway	9	3.26×10^{-16}
hsa04933	AGE–RAGE signalling pathway in diabetic complications	10	3.54×10^{-17}
hsa04932	Non-alcoholic fatty liver disease	12	1.06×10^{-19}

hsa03320	PPAR signalling pathway	3	1.70×10^{-4}

Table 3. Top enriched KEGG pathways associated with the common therapeutic targets.

Discussion

KEGG pathway enrichment analysis demonstrated that the common therapeutic targets are involved in multiple interconnected pathways that regulate glucose metabolism, insulin sensitivity, lipid metabolism, inflammation, and cellular energy balance. Significant enrichment of the Type II diabetes mellitus, insulin resistance, insulin signalling, AMPK signalling, and FoxO signalling pathways suggests that berberine may exert its antidiabetic activity through the coordinated modulation of several molecular mechanisms rather than acting on a single target.

The enrichment of the AMPK signalling pathway supports previous studies reporting that berberine activates AMPK to enhance glucose uptake, inhibit hepatic gluconeogenesis, and improve insulin sensitivity. Likewise, enrichment of the Insulin signalling pathway, mTOR signalling pathway, and Adipocytokine signalling pathway indicates that the identified targets participate in maintaining metabolic homeostasis and regulating cellular responses to insulin.

Although the PPAR signalling pathway showed comparatively lower statistical enrichment than several other pathways, it was considered biologically significant because it contains PPARG, a ligand-activated nuclear receptor that plays a central role in regulating glucose homeostasis, lipid metabolism, adipocyte differentiation, and insulin sensitivity. Since PPARG is a clinically validated therapeutic target for Type 2 Diabetes Mellitus and is directly associated with the pharmacological mechanism of several approved antidiabetic drugs, PPARG was selected as the target protein for subsequent molecular docking studies with berberine.

4.5 Molecular Docking Analysis

Molecular docking was performed to investigate the binding interaction between berberine and Peroxisome Proliferator-Activated Receptor Gamma (PPARG) using both the CB-Dock2 web server and AutoDock Vina (Version 1.2.x). CB-Dock2 was initially employed for automatic binding cavity identification and preliminary docking analysis, while AutoDock Vina was subsequently used to perform docking simulations using the prepared PDBQT files, allowing independent evaluation of the binding affinity and ligand binding poses.

Multiple docking conformations were generated and ranked according to their predicted binding energies (kcal/mol). The docking pose with the lowest binding energy was selected for further protein–ligand interaction analysis. Both docking approaches predicted a favourable interaction between berberine and PPARG, indicating the formation of a stable protein–ligand complex within the receptor binding pocket.

The molecular docking results are summarized in Table 4, and the predicted binding pose of the berberine–PPARG complex is shown in Figure 3 and 4.

Parameter	AutoDock Vina	CB-Dock2
Target protein	PPARG	PPARG
Ligand	Berberine	Berberine
Best binding affinity (kcal/mol)	-10.3	-9.5
Predicted binding cavity	Grid-defined active site	Predicted Cavity 1
Cavity volume(Å^3)	Not applicable	1518

Table 4. Molecular docking results of berberine with PPARG obtained using AutoDock Vina and CB-Dock2

Figure 3. Three-dimensional binding pose of berberine in the active site of PPARG

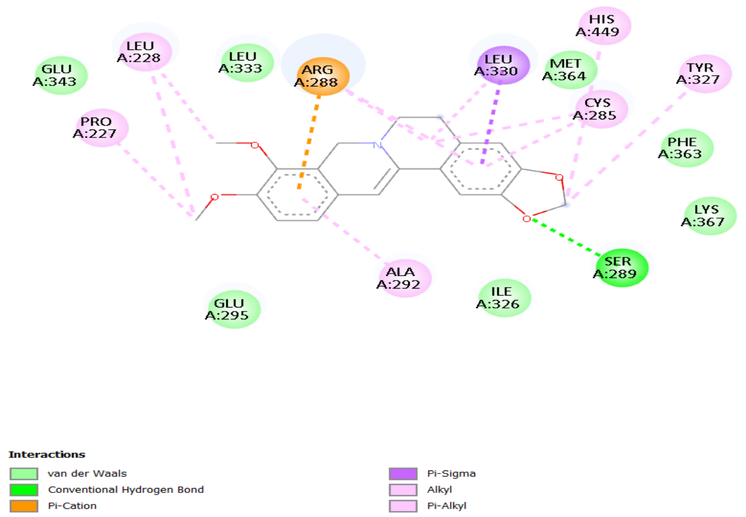

Figure 4. Two-dimensional protein–ligand interaction diagram

4.6 ADME Prediction Using SwissADME

The pharmacokinetic properties, drug-likeness, and medicinal chemistry profile of berberine were evaluated using the SwissADME web server to assess its potential as an orally active therapeutic agent. The analysis included physicochemical properties, lipophilicity, water solubility, gastrointestinal absorption, blood–brain barrier permeability, drug-likeness, and medicinal chemistry parameters.

SwissADME predicted that berberine possesses favourable physicochemical properties, with a molecular weight of 336.36 g/mol, Topological Polar Surface Area (TPSA) of 40.80 Å^2, and a consensus Log P value of 2.82, indicating moderate lipophilicity. Water solubility analysis classified berberine as moderately soluble according to the ESOL, Ali, and SILICOS-IT prediction models.

Pharmacokinetic analysis predicted high gastrointestinal absorption, suggesting good oral bioavailability, while berberine was also predicted to be blood–brain barrier permeable and a P-glycoprotein substrate. Drug-likeness evaluation indicated that berberine satisfied Lipinski's Rule of Five, Ghose, Veber, Egan, and Muegge filters without any violations. Furthermore, the compound exhibited a bioavailability score of 0.55, indicating favourable oral bioavailability.

Medicinal chemistry analysis revealed no PAINS alerts, suggesting a low probability of assay interference. However, one Brenk structural alert (quaternary nitrogen) and one lead-likeness violation (XLOGP3 > 3.5) were identified, indicating that structural optimization may further improve its pharmaceutical properties. Overall, the SwissADME analysis suggests that berberine possesses favourable pharmacokinetic and drug-likeness characteristics, supporting its potential for further development as a therapeutic agent against Type 2 Diabetes Mellitus.

The predicted ADMET and drug-likeness properties of berberine are summarized in Table 5, while the SwissADME bioavailability radar and pharmacokinetic profile are presented in Figure 5.

Property	**Predicted Value**
SMILES	COC1=C(C2=C[N+]3=C(C=C2C=C1)C4=CC5=C(C=C4CC3)OCO5)OC
Molecular Formula	$C_{20}H_{18}NO_4^+$
Molecular Weight	336.36 g/mol
Heavy Atoms	25
Aromatic Heavy Atoms	16
Fraction Csp^3	0.25
Rotatable Bonds	2
Hydrogen Bond Acceptors	4
Hydrogen Bond Donors	0
Molar Refractivity	94.87
TPSA	40.80 $Å^2$
Consensus Log P	2.82
Water Solubility	Moderately soluble
GI Absorption	High

BBB Permeability	Yes
P-glycoprotein Substrate	Yes
Skin Permeation (Log Kp)	-5.78 cm/s
Lipinski Rule of Five	Pass (0 violations)
Ghose Filter	Pass
Veber Filter	Pass
Egan Filter	Pass
Muegge Filter	Pass
Bioavailability Score	0.55
PAINS Alerts	0
Brenk Alerts	1 (Quaternary nitrogen)
Lead-likeness	No (1 violation: XLOGP3 > 3.5)
Synthetic Accessibility	3.14

Table 5. Physicochemical properties, pharmacokinetic parameters, drug-likeness, and medicinal chemistry profile of berberine predicted using the SwissADME web server

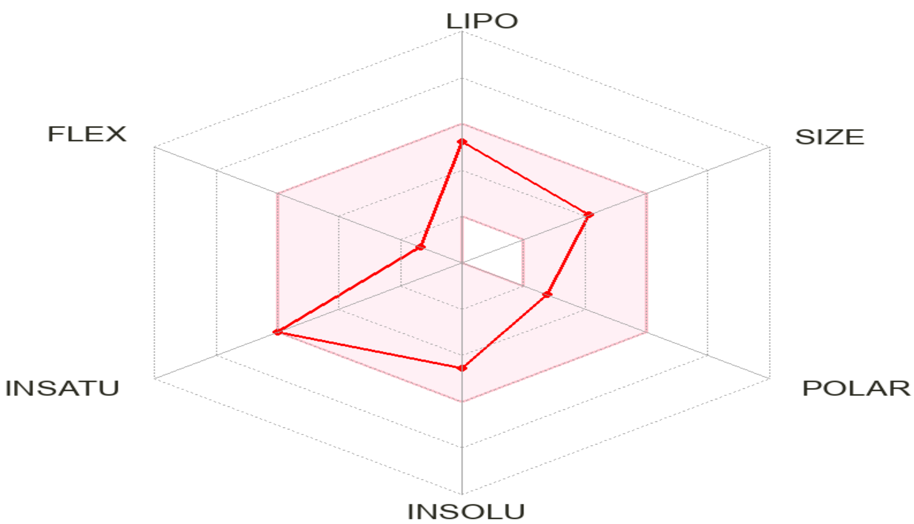

Figure 5. ***SwissADME bioavailability radar of berberine illustrating its predicted physicochemical properties associated with oral bioavailability. The pink region represents the optimal range for oral drug candidates, while the red polygon corresponds to the physicochemical profile of berberine***

Discussion

The SwissADME analysis indicated that berberine possesses favourable physicochemical and pharmacokinetic characteristics that support its potential as an orally active therapeutic compound. The compound exhibited high gastrointestinal absorption, suggesting efficient oral uptake, while its consensus Log P value (2.82) and TPSA (40.80 $Å^2$) indicate a balanced lipophilicity and polarity favourable for membrane permeability. The bioavailability radar further demonstrated that most of the physicochemical properties of berberine fall within the optimal range for oral drug candidates.

Berberine complied with Lipinski's Rule of Five, as well as the Ghose, Veber, Egan, and Muegge drug-likeness filters, indicating good drug-like properties. The predicted bioavailability score of 0.55 further supports its suitability for oral administration. Although berberine was predicted to be a P-glycoprotein substrate and exhibited one Brenk structural alert (quaternary nitrogen) together with one lead-likeness violation, these findings do not preclude its therapeutic potential but suggest that structural optimization could further improve its pharmaceutical properties.

Overall, the SwissADME results indicate that berberine possesses an acceptable ADMET profile and favourable drug-likeness characteristics, supporting its continued investigation as a potential therapeutic agent for the management of Type 2 Diabetes Mellitus.

4.7 Toxicity Prediction

The toxicity profile of berberine was evaluated using the ProTox-3.0 web server to predict its potential organ toxicity, toxicological endpoints, metabolism, and molecular initiating events. ProTox-3.0 employs machine learning-based models to estimate toxicity based on the chemical structure of a compound, providing an initial assessment of its safety profile.

The analysis predicted that berberine is inactive for hepatotoxicity, nephrotoxicity, cardiotoxicity, clinical toxicity, and nutritional toxicity, suggesting a relatively favourable safety profile with respect to these parameters. However, neurotoxicity and respiratory toxicity were predicted to be active. Among the toxicity endpoints, berberine showed predicted activity for carcinogenicity, immunotoxicity, mutagenicity, cytotoxicity, blood–brain barrier permeability, and ecotoxicity.

Metabolism prediction indicated that berberine may interact with several cytochrome P450 enzymes, including CYP1A2, CYP2C19, CYP2D6, and CYP3A4, whereas CYP2C9 and CYP2E1 were predicted to be inactive. Furthermore, toxicity pathway analysis suggested activation of the Aryl Hydrocarbon Receptor (AhR) and the mitochondrial membrane potential (MMP) pathway, while most other nuclear receptor and stress response pathways were predicted to remain inactive.

The predicted toxicity profile of berberine is summarized in Table 6. The ProTox-3.0 toxicity radar and network visualization are presented in Figure 6.

Category	Parameter	Predicted Value
General Toxicity	Predicted LD_{50}	200 mg/kg
	Toxicity Class	Class III (Toxic if swallowed)
	Prediction Accuracy	67.38%
	Average Similarity	50.04%
Physicochemical Properties	Molecular Weight	336.36 g/mol

	Hydrogen Bond Acceptors	4
	Hydrogen Bond Donors	0
	Rotatable Bonds	2
	Molecular Refractivity	94.87
	TPSA	40.80 Å^2
	Log P	3.10
Organ Toxicity	Hepatotoxicity	Inactive
	Neurotoxicity	Active
	Nephrotoxicity	Inactive
	Respiratory Toxicity	Active
	Cardiotoxicity	Inactive
Toxicity Endpoints	Carcinogenicity	Active
	Immunotoxicity	Active
	Mutagenicity	Active
	Cytotoxicity	Active
	Blood–Brain Barrier	Active
	Ecotoxicity	Active
	Clinical Toxicity	Inactive
	Nutritional Toxicity	Inactive

Metabolism	CYP1A2	Active
	CYP2C19	Active
	CYP2C9	Inactive
	CYP2D6	Active
	CYP3A4	Active
	CYP2E1	Inactive

Table 6. Predicted acute toxicity, organ toxicity, toxicity endpoints, and metabolic interactions of berberine obtained using the ProTox-3.0 web serve

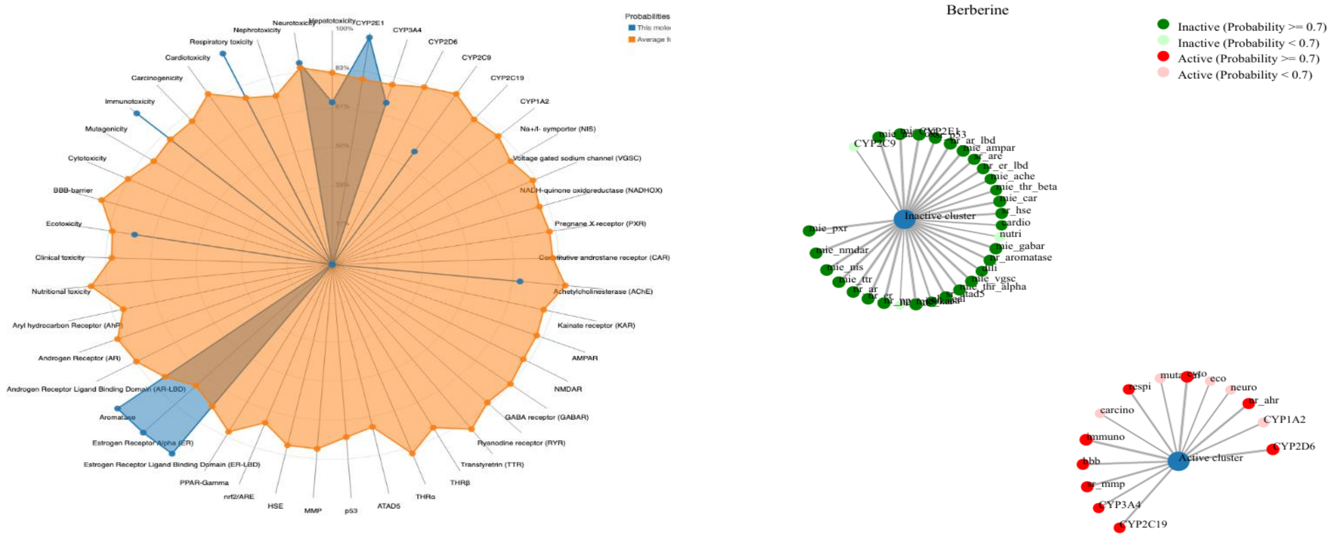

Figure 6. ***ProTox-3.0 toxicity radar and network visualization illustrating the predicted toxicity profile and toxicity-associated targets of berberine***

Discussion

The ProTox-3.0 toxicity prediction indicated that berberine possesses a moderate acute toxicity profile, with a predicted oral LD_{50} of 200 mg/kg corresponding to Toxicity Class III under the Globally Harmonized System (GHS). The compound was predicted to be inactive for hepatotoxicity, nephrotoxicity, cardiotoxicity, clinical toxicity, and nutritional toxicity, suggesting a relatively favourable safety profile with respect to these endpoints.

However, berberine was predicted to exhibit neurotoxicity and respiratory toxicity, together with positive predictions for carcinogenicity, mutagenicity, immunotoxicity, and cytotoxicity. In addition, the metabolism analysis suggested potential interactions with CYP1A2, CYP2C19, CYP2D6, and CYP3A4, indicating that berberine may undergo metabolism through multiple cytochrome P450 enzymes. These computational predictions provide valuable preliminary information regarding the safety profile of berberine but should be interpreted with caution, as they are based on machine learning models and require validation through *in vitro*, *in vivo*, and clinical studies. Overall, the toxicity assessment suggests that although berberine demonstrates promising therapeutic potential, further experimental evaluation is necessary to establish its safety and therapeutic window.

5. Discussion

The present study employed an integrated computational approach combining network pharmacology, molecular docking, ADME prediction, and toxicity assessment to investigate the therapeutic potential of berberine against Type 2 Diabetes Mellitus (T2DM). This strategy enabled the identification of key molecular targets and biological pathways that may contribute to the antidiabetic effects of berberine.

Target prediction and disease-target screening identified 19 overlapping genes between berberine-associated targets and T2DM-related genes. The highly interconnected protein–protein interaction network obtained from STRING suggests that these proteins function cooperatively in regulating glucose metabolism, insulin signalling, inflammation, and cellular homeostasis. The significant PPI enrichment ($p < 1.0 \times 10^{-16}$) further supports the biological relevance of these interactions.

KEGG pathway enrichment analysis revealed that the identified targets are mainly involved in pathways associated with insulin resistance, insulin signalling, Type II diabetes mellitus, AMPK signalling, FoxO signalling, adipocytokine signalling, PPAR signalling, and non-alcoholic fatty liver disease. These pathways are well established in the pathogenesis of T2DM and play essential roles in regulating glucose uptake, insulin sensitivity, lipid metabolism, and inflammatory responses. The enrichment of these pathways suggests that berberine may exert its therapeutic effects through a multi-target and multi-pathway mechanism, which is a characteristic feature of many natural products.

Molecular docking demonstrated a strong interaction between berberine and PPARG, yielding a binding affinity of −10.3 kcal/mol using AutoDock Vina. CB-Dock2 also predicted favourable binding with a best docking score of −9.5 kcal/mol, supporting the stability of the protein–ligand complex. Since PPARG is a major regulator of glucose and lipid metabolism, these findings indicate that berberine may improve insulin sensitivity through modulation of PPARG activity.

The SwissADME analysis showed that berberine possesses favourable physicochemical and pharmacokinetic characteristics, including high gastrointestinal absorption, moderate lipophilicity, and compliance with major drug-likeness rules. Although the compound was predicted to be a P-glycoprotein substrate and exhibited one Brenk alert together with one lead-likeness violation, its overall ADMET profile supports its suitability for further drug development.

ProTox-3.0 predicted an oral LD_{50} of 200 mg/kg (Toxicity Class III), indicating moderate acute toxicity. The compound was predicted to be inactive for hepatotoxicity, nephrotoxicity, and cardiotoxicity, but active predictions for neurotoxicity, respiratory toxicity, mutagenicity, carcinogenicity, immunotoxicity, and cytotoxicity highlight the importance of careful dose optimization and experimental validation. These predictions should be interpreted as preliminary computational assessments rather than definitive evidence of toxicity.

Overall, the results of this study suggest that berberine has the potential to act as a multi-target therapeutic agent for T2DM by modulating key proteins and signalling pathways involved in glucose metabolism and insulin resistance. Nevertheless, the present findings are based entirely on computational analyses. Therefore, *in vitro*, *in vivo*, and clinical investigations are necessary to validate the predicted molecular interactions, pharmacokinetic behaviour, and safety profile before therapeutic application.

6. Conclusion

The present study employed an integrated network pharmacology and molecular docking approach to investigate the therapeutic potential of berberine against Type 2 Diabetes Mellitus (T2DM). A total of 19 common target genes were identified between berberine-associated targets and T2DM-related genes, indicating that berberine exerts its therapeutic effects through a multi-target mechanism.

Protein–protein interaction analysis and KEGG pathway enrichment demonstrated that these targets are significantly involved in important biological pathways related to insulin signalling, insulin resistance, AMPK signalling, FoxO signalling, PPAR signalling, adipocytokine signalling, and glucose metabolism**.** These findings suggest that berberine may regulate multiple signalling pathways associated with the pathogenesis of T2DM.

Molecular docking analysis revealed a strong binding interaction between berberine and Peroxisome Proliferator-Activated Receptor Gamma (PPARG), with AutoDock Vina predicting a binding affinity of −10.3 kcal/mol, while CB-Dock2 predicted a favourable binding score of −9.5 kcal/mol. These results support the potential of berberine to modulate PPARG activity, thereby improving insulin sensitivity and glucose homeostasis.

SwissADME analysis indicated that berberine possesses favourable physicochemical properties, acceptable pharmacokinetic characteristics, and good drug-likeness, supporting its potential as an orally active therapeutic candidate. Although ProTox-3.0 predicted moderate acute toxicity (LD_{50} = 200 mg/kg, Toxicity Class III) and identified several toxicity endpoints that warrant further investigation, these predictions represent preliminary computational assessments and require experimental validation.

Overall, the findings of this study suggest that berberine is a promising multi-target natural compound with potential therapeutic value in the management of Type 2 Diabetes Mellitus. However, further i*n vitro*, *in vivo*, and clinical studies are necessary to validate the predicted molecular mechanisms, pharmacokinetic behaviour, efficacy, and safety before its clinical application.

7. Future Scope

Although the present study provides valuable computational insights into the therapeutic potential of berberine against Type 2 Diabetes Mellitus, further investigations are necessary to validate these findings. Future research should focus on:

- Experimental validation of the predicted target proteins through *in vitro* biochemical and cell-based assays.
- *In vivo* studies to evaluate the antidiabetic efficacy and pharmacological effects of berberine in suitable animal models.
- Investigation of the molecular mechanisms underlying the regulation of insulin signalling, glucose metabolism, and inflammatory pathways by berberine.
- Optimization of berberine derivatives to improve bioavailability, pharmacokinetic properties, and therapeutic efficacy.
- Clinical studies to evaluate the safety, efficacy, and therapeutic potential of berberine in patients with Type 2 Diabetes Mellitus.
- Application of artificial intelligence and molecular dynamics simulations to further explore protein–ligand interactions and facilitate rational drug design.

These studies will contribute to a better understanding of the pharmacological potential of berberine and support its development as a novel therapeutic agent for the treatment of Type 2 Diabetes Mellitus.

8. Limitations

The present study is based entirely on computational approaches, and therefore several limitations should be acknowledged:

- The predicted molecular targets and biological pathways are based on publicly available databases and computational algorithms.
- Molecular docking provides only a theoretical estimation of protein–ligand binding affinity and does not fully represent biological conditions.
- ADMET and toxicity predictions are based on machine learning models and require experimental confirmation.
- Protein flexibility, solvent effects, and dynamic molecular interactions were not considered in the docking analysis.
- Experimental validation through *in vitro*, *in vivo*, and clinical studies was beyond the scope of the present work.

Despite these limitations, the integrated computational approach provides valuable preliminary evidence supporting the therapeutic potential of berberine against Type 2 Diabetes Mellitus.

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